Form A

1. (a) There are various questions, but here are two reasonable ones

I. Are the 12 months equally likely as birth months?

$$H_0: p_1 = p_2 = p_3 = \cdots = p_{12} = \frac{1}{12}$$

II. Do birth months occur in proportion to the number of days?

$$H_0: p_1 = p_3 = p_5 = p_7 = p_8 = p_{10} = p_{12} = \frac{31}{365}, \quad p_2 = \frac{28}{365},$$
$$p_4 = p_6 = p_9 = p_{11} = \frac{30}{365}$$

No statement of alternative hypothesis is required, but a concise statement can always(?) be H_a: At least one p_i is not as proposed.

(b) There is much less flexibility in responses here.

Is there an association between the categorical variables "birth month" and "hemisphere"?

H_0: The variables "birth month" and "hemisphere" are independent.

2. (a) You should check that the expected frequencies in all 12 cells are at least 5.

(b) 1 - pchisq(3.178, df = 6)